
Project and Case Study Fields

1) Prediction from Networks (Graph Data)

Methods / techniques

- Graph ML baselines: node2vec / DeepWalk, GraphSAGE, GAT, GCN
- Tabular baselines on engineered graph features (degree, centrality, motifs)
- Link prediction + node/graph classification pipelines

Possible novelties

- **Representative subgraph sampling** (your Black Hole / SOCIAL) for faster training
- **Graph shift robustness**: test when network structure changes over time
- **Explainable subgraph rationales** for decisions

2) Graph Neural Networks for Real-World Problems (GNN Benchmarking)

Methods / techniques

- Strong baselines: XGBoost/LightGBM on graph statistics, MLP on embeddings
- GNNs: GraphSAGE/GAT/GIN; graph pooling if graph-level task
- Evaluation: multiple splits, OOD split, ablations (features/edges/labels)

Possible novelties

- **“GNN vs non-GNN” truth test**: when do GNNs really win?
- **Cost-aware benchmarking** (accuracy vs training time vs memory)
- **Generalization study** across datasets/domains (science/business/health)

3) Smart Sampling (Coresets) for Faster Training

Methods / techniques

- Coreset selection: k-center greedy, facility location, herding, stratified sampling
- Embedding-based selection (use a pretrained encoder, then select representatives)
- Your metaheuristics: **Black Hole Strategy** / **LEA** / **SOCIAL** for subset selection

Possible novelties

- A **new coreset objective**: diversity + uncertainty + class balance
 - Show coreset benefits under **limited compute** (edge / laptop setting)
 - **Cross-domain coreset**: subset chosen on one dataset transfers to another
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4) Better Features for Molecules (AI Representations)

Methods / techniques

- Molecular encoders: ECFP/Morgan fingerprints, RDKit descriptors
- Graph models: MPNN, GIN, GAT
- Self-supervised learning: contrastive learning on graphs, masked-atom prediction

Possible novelties

- **Dual-view fusion** (SMILES + graph) with simple gating/attention
 - **Uncertainty-calibrated screening** (rank candidates with confidence)
 - Robustness tests: stereochemistry noise, missing atoms/bonds, scaffold split
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5) AI for Reading Papers (Text Mining + Info Extraction)

Methods / techniques

- NLP pipelines: NER, relation extraction, sentence classification
- LLM + RAG with evidence highlighting; citation-quality scoring
- Weak supervision (labeling functions), distant supervision from databases

Possible novelties

- **Hallucination-resistant extraction**: only accept claims with evidence spans
 - **Structured “property cards”** auto-generated from papers (data cards style)
 - Domain adaptation: how to transfer extraction from one field to another
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6) AI for Images: Quality Control or Medical/Scientific Imaging

Methods / techniques

- CNN/ViT classification, segmentation (U-Net), anomaly detection (autoencoders)
- Self-supervised pretraining (SimCLR/BYOL-style) for small labeled sets

- Evaluation: Dice/IoU (segmentation), AUROC (anomaly), calibration

Possible novelties

- **Few-shot defect detection** with strong baselines + ablations
 - **Uncertainty + abstention** (“refer to human” mode)
 - **Shift tests**: new camera/device/lighting → robustness study
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7) AI-Assisted Design & Optimization (AutoML + Metaheuristics)

Methods / techniques

- Hyperparameter optimization: random search, Bayesian optimization, Hyperband
- Feature selection: L1, mutual information, wrappers
- Your algorithms: **LEA** / **SOCIAL** / **Black Hole** as search engines

Possible novelties

- **Multi-objective AutoML** (accuracy + latency + interpretability)
 - A **fair comparison protocol** for metaheuristics (budget-matched, seeds, stats)
 - Hybrid: metaheuristic initializes → Bayesian optimization refines (or reverse)
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8) AI for Clean Energy (Forecasting + Optimization)

Methods / techniques

- Forecasting: XGBoost, Prophet-style baselines, LSTM/TCN, TFT (if advanced)
- Optimization: scenario analysis, multi-objective trade-offs, metaheuristics
- Explainability for time-series: SHAP on lag features, error decomposition

Possible novelties

- **Probabilistic forecasting** + decision-making (optimize using prediction intervals)
 - **Concept drift detection** for changing markets/weather patterns
 - “Small data” strategy: transfer learning or self-supervised time-series features
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9) AI in Drug Discovery (Screening Pipelines)

Methods / techniques

- QSAR baselines: RF/XGBoost on fingerprints, graph models (MPNN/GIN)
- Splits: scaffold split (critical), temporal split if possible
- Active learning: pick next compounds to label (uncertainty/diversity)

Possible novelties

- **Active learning with diversity constraints** (your coreset angle)
- **Ranking quality** focus (top-K hit rate) instead of only RMSE/AUROC
- Robustness: **OOD scaffold generalization** + calibration

10) Reliable AI (Uncertainty, Calibration, “I Don’t Know”)

Methods / techniques

- Calibration: temperature scaling, isotonic regression, Dirichlet calibration
- Selective prediction: confidence thresholds, conformal prediction
- Robustness: stress tests, synthetic noise, subgroup performance

Possible novelties

- **Cost-sensitive abstention** (optimize threshold for business/health costs)
- **Conformal prediction** for tabular/time-series pipelines in applied settings
- Reliability benchmark: compare calibration across model families + drift settings